

Interview-Style Questions Based on Conditional Statements

1. Check Even or Odd

Question: Determine whether a number is even or odd. **Explanation:** A number is even if it is divisible by 2. Otherwise, it's odd. - **Input:** Number = 6 - **Output:** Even number

Program:

```
n=6
```

```
if n%2==0:
```

```
    print("Even number")
```

```
else:
```

```
    print("Odd number")
```

Output:

Even number

2. Divisible by 5 but Not by 10

Question: Check if a number is divisible by 5 but not by 10. **Explanation:** Use modulo (%) to check if the number % 5 == 0 and number % 10 != 0. - **Input:** Number = 25 - **Output:** Satisfy

Program:

```
n=25
```

```
if n%5==0 and n%10!=0:
```

```
    print("Satisfy")
```

```
else:
```

```
    print("Not Satisfy")
```

Output:

Satisfy

3. Biggest Among Two Numbers

Question: Find the biggest number among two. **Explanation:** Use comparison operators (>) to check which number is greater. - **Input:** A = 4, B = 7 - **Output:** Biggest is: 7

Program:

```
A=4
```

```
B=7
```

if A>B:

```
    print("Biggest number:",A)
```

else:

```
    print("Biggest number:",B)
```

Output:

Biggest number:7

4. Smallest Among Two Numbers

Question: Find the smallest number among two. **Explanation:** Use comparison operators (<) to find the smaller value. - **Input:** A = 4, B = 7 - **Output:** Smallest is: 4

Program:

```
A=4
```

```
B=7
```

if A<B:

```
    print("Smallest number: ", A)
```

else:

```
    print("Smallest number :", B)
```

Output:

Smallest number:4

5. Divisible by 2, 3, and 6

Question: Check if a number is divisible by 2, 3, and 6. **Explanation:** If a number is divisible by both 2 and 3, it is also divisible by 6. - **Input:** Number = 18 - **Output:** Satisfy

Program:

```
n=18
```

```
if n%2==0 and n%3==0 and n%6==0:
```

```
    print("Satisfy")
```

else:

```
    print("Not Satisfy")
```

Output:

Satisfy

6. Voting Eligibility

Question: Check if a person is eligible to vote (age ≥ 18). **Explanation:** A person is eligible to vote if their age is 18 or above. - **Input:** Age = 19 - **Output:** Eligible to vote

Program:

```
age=19
```

```
if age<=18:
```

```
    print("Eligible to vote")
```

```
else:
```

```
    print("Not Eligible to vote")
```

output:

Eligible to vote

7. Student Pass/Fail Based on All Subjects ≥ 35

Question: Check if a student passed all subjects (maths, physics, chemistry). **Explanation:** Student passes only if marks in all subjects are 35 or more. - **Input:** Maths = 40, Physics = 36, Chemistry = 30 - **Output:** Fail

Program:

```
Maths=40
```

```
Physics=36
```

```
Chemistry=30
```

```
if Maths<=35 and Physics<=35 and Chemistry<=35:
```

```
    print("Pass")
```

```
else:
```

```
    print("Fail")
```

Output:

Fail

8. Student Pass if Passed Any One Subject (≥ 35)

Question: Check if the student passed at least one subject. **Explanation:** Use logical OR to check if any one subject has marks ≥ 35 . - **Input:** Maths = 20, Physics = 38, Chemistry = 25 - **Output:** Pass

Program:

Maths=20

Physics=38

Chemistry=25

if Maths>=35 or Physics>=35 or Chemistry>=35:

 print("Pass")

else:

 print("Fail")

Output:

Pass

9. Student Pass if Passed Any Two Subjects

Question: Check if the student passed any two out of three subjects. **Explanation:** Use a counter or logical conditions to verify two subjects >= 35. - **Input:** Maths = 40, Physics = 20, Chemistry = 36 -

Output: Pass

Program:

maths =40

physics =20

chemistry =36

count = 0

if maths >= 35:

 count = count + 1

if physics >= 35:

 count = count + 1

if chemistry >= 35:

 count = count + 1

if count >= 2:

 print("Pass")

else:

 print("Fail")

Output:

Pass

10. Biggest Among Three Numbers

Question: Find the biggest number among three. **Explanation:** Compare each pair of numbers using if-else conditions. - **Input:** A = 7, B = 4, C = 9 - **Output:** Biggest is: 9

Program:

A=7

B=4

C=9

if A>B and A>C:

 print("Biggest is:",A)

elif B>A and B>C:

 print("Biggest is:",B)

else:

 print("Biggest is:",C)

Output:

Biggest is:9

11. Smallest Among Three Numbers

Question: Find the smallest number among three. **Explanation:** Use comparison logic to determine the minimum value. - **Input:** A = 7, B = 4, C = 9 - **Output:** Smallest is: 4

Program:

A=7

B=4

C=9

if A<B and A<C:

 print("Smallest is:",A)

elif B<A and B<C:

 print("Smallest is:",B)

else:

 print("Smallest is:",C)

Output:

Smallest is:4

12. Perfect Square or Not

Question: Check if a number is a perfect square. **Explanation:** A number is a perfect square if the square of its square root equals the number. - **Input:** Number = 49 - **Output:** Perfect square

Program:

```
n=49
root=int(n**0.5)
if root*root==n:
    print("Perfect Square")
else:
    print("Not Perfect Square")
```

Output:

Perfect Square

13. Cars Required for Members (Max 5 per car)

Question: Calculate how many cars are needed for a given number of people. **Explanation:** Divide total people by 5 and round up using ceiling logic. - **Input:** Members = 17 - **Output:** Cars needed = 4

Program:

```
members = 17
if members % 5 == 0:
    cars = members // 5
else:
    cars = (members // 5) + 1

print("Cars needed =", cars)
```

Output:

Cars needed = 4

14. Second Biggest Among Three Numbers

Question: Find the second largest number among three inputs. **Explanation:** Use sorting or nested conditions to find the second largest value. - **Input:** A = 10, B = 25, C = 18 - **Output:** Second biggest: 18

Program:

A = 10

B = 25

C = 18

if (A>B and A<C) or (A<B and A>C):

 print("Second largest is:",A)

elif (B>A and B<C) or (B<A and B>C):

 print("Second largest is:",B)

else:

 print("Second largest is:",C)

Output:

Second largest is:18

15. Leap Year or Not

Question: Check if a given year is a leap year. **Explanation:** A year is a leap year if it is divisible by 4, and (not divisible by 100 unless divisible by 400). - **Input:** Year = 2024 - **Output:** Leap year

Program:

Year = 2024

if Year % 400 == 0 or (Year % 4 == 0 and Year % 100 != 0):

 print("Leap Year")

else:

 print("Not Leap Year")

Output:

Leap Year